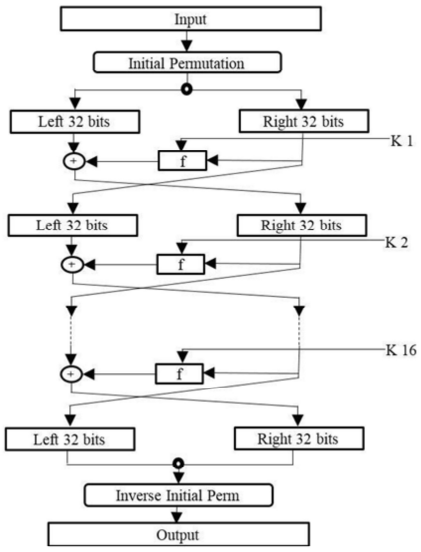
# Data Encryption Standard




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Reference https://searchsecurity.techtarget.com/definition/Data-Encryption-Standard

The Data Encryption Standard (DES) is an outdated symmetric-key method of data encryption.
DES works by using the same key to encrypt and decrypt a message, so both the sender and the receiver must know and use the same private key. Once the go-to, symmetric-key algorithm for the encryption of electronic data, DES has been superseded by the more secure Advanced Encryption Standard (AES) algorithm.
Originally designed by researchers at IBM in the early 1970s, DES was adopted by the U.S. government as an official Federal Information Processing Standard (FIPS) in 1977 for the encryption of commercial and sensitive yet unclassified government computer data. It was the first encryption algorithm approved by the U.S. government for public disclosure. This ensured that DES was quickly adopted by industries such as financial services, where the need for strong encryption is high. The simplicity of DES also saw it used in a wide variety of embedded systems, smart cards, SIM cards and network devices requiring encryption like modems, set-top boxes and routers.

**DES key length and brute-force attacks**

The Data Encryption Standard is a block cipher, meaning a cryptographic key and algorithm are applied to a block of data simultaneously rather than one bit at a time. To encrypt a plaintext message, DES groups it into 64-bit blocks. Each block is enciphered using the secret key into a 64-bit ciphertext by means of permutation and substitution. The process involves 16 rounds and can run in four different modes, encrypting blocks individually or making each cipher block dependent on all the previous blocks. Decryption is simply the inverse of encryption, following the same steps but reversing the order in which the keys are applied. For any cipher, the most basic method of attack is brute force, which involves trying each key until you find the right one. The length of the key determines the number of possible keys -- and hence the feasibility -- of this type of attack. DES uses a 64-bit key, but eight of those bits are used for parity checks, effectively limiting the key to 56-bits. Hence, it would take a maximum of 2^56, or 72,057,594,037,927,936, attempts to find the correct key.

Even though few messages encrypted using DES encryption are likely to be subjected to this kind of code-breaking effort, many security experts felt the 56-bit key length was inadequate even before DES was adopted as a standard. (There have always been suspicions that interference from the NSA weakened IBM's original algorithm). Even so, DES remained a trusted and widely used encryption algorithm through the mid-1990s. However, in 1998, a computer built by the Electronic Frontier Foundation (EFF) decrypted a DES-encoded message in 56 hours. By harnessing the power of thousands of networked computers, the following year EFF cut the decryption time to 22 hours.

Apart from providing backwards compatibility in some instances, reliance today upon DES for data confidentiality is a serious security design error in any computer system and should be avoided. There are much more secure algorithms available, such as AES. Much like a cheap suitcase lock, DES will keep the contents safe from honest people, but it won't stop a determined thief.

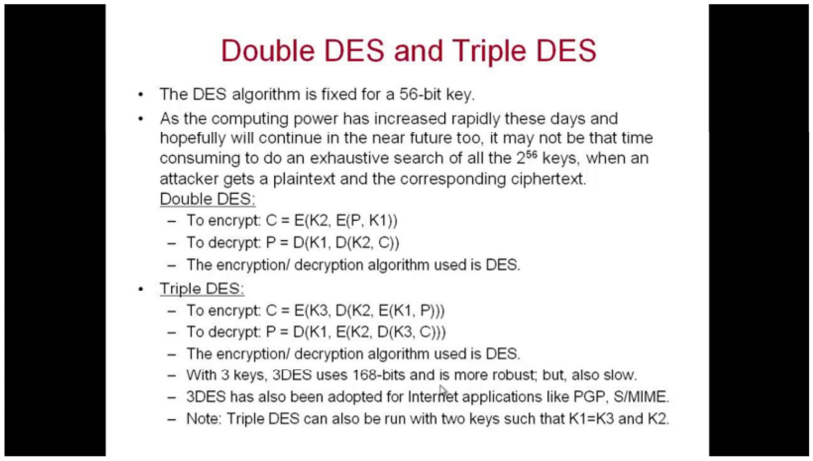
**Successors to DES**

Encryption strength is directly tied to key size, and 56-bit key lengths have become too small relative to the processing power of modern computers. So in 1997, the National Institute of Standards and Technology (NIST) announced an initiative to choose a successor to DES; in 2001, it selected the Advanced Encryption Standard as a replacement. The Data Encryption Standard (FIPS 46-3) was officially withdrawn in May 2005, though Triple DES (3DES) is approved through 2030 for sensitive government information. 3DES performs three iterations of the DES algorithm; if keying option number one is chosen, a different key is used each time to increase the key length to 168 bits. However, due to the likelihood of a meet-in-the-middle attack, the effective security it provides is only 112 bits. 3DES encryption is obviously slower than plain DES.

**Legacy of DES**

Despite having reached the end of its useful life, the arrival of the Data Encryption Standard served to promote the study of cryptography and the development of new encryption algorithms. Until DES, cryptography was a dark art confined to the realms of military and government intelligence organizations. The open nature of DES meant academics, mathematicians and anyone interested in security could study how the algorithm worked and try to crack it. As with any popular and challenging puzzle, a craze -- or in this case, a whole industry -- was born.

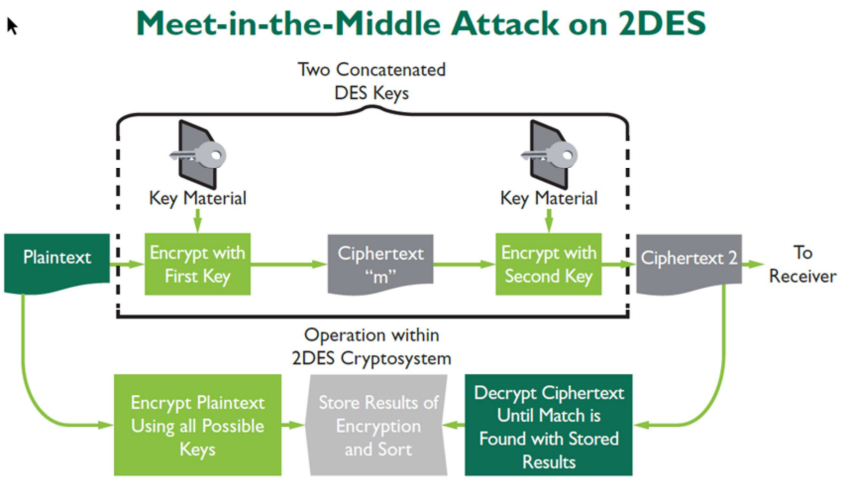
# Double DES and Triple DES




As we've seen, the main problem with DES is that the key is too short to provide adequate protection against brute force attacks. Increasing the key length is an effective defense against a brute force attack. Ways to improve the DES algorithm's resistance to a brute force attack have been developed by the industry. These efforts are referred to as Double DES and Triple DES.

Double-DES refers to the use of two DES encryptions with two separate keys, effectively doubling the size of the DES key from 56 bits to 112 bits. This dramatic increase in key size much more than doubles the strength of the cipher. Each increase of a single bit effectively doubles the number of keys in the keyspace. This means that a 57-bit key space is twice as large as a 56-bit key space. A 58-bit key is four times as big, etc. This would seem like a vast improvement in strength against brute force; however, there is an attack on Double-DES that reduces its effective number of keys to about the same number in DES. This attack is known as the meet-in-the-middle attack, and it reduces the strength of Double-DES to almost the same as DES.

# Meet Man in the Middle Attack on DES

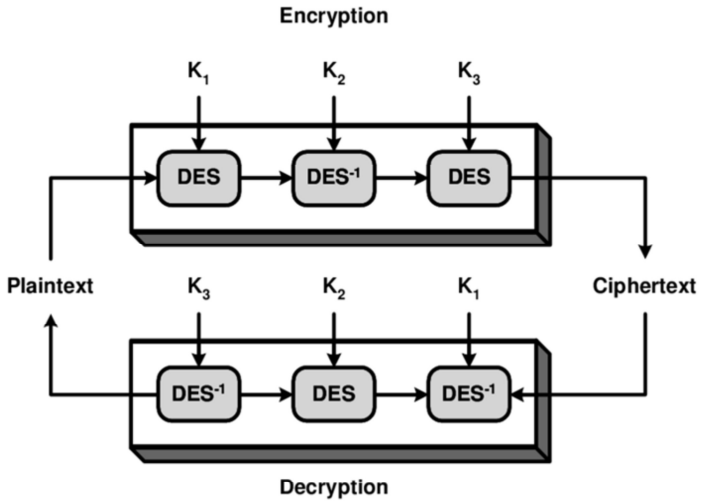



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3

A very effective attack against double DES is based on doing a brute force attack against known plaintext. This attack is known as the meet man-in-the-middle attack. The attacker would encrypt the plaintext using all possible keys and create a table containing all possible results. This intermediate cipher is referred to as “m” for this discussion. This would mean encrypting using all 2 to the power of 56 possible keys. The table would then be sorted according to the values of “m.” The attacker would then decrypt the ciphertext using all possible keys until he found a match with the value of “m.” This would result in a true strength of double DES of approximately 2 to the power of 57 (twice the strength of DES but not strong enough to be considered effective) instead of the 2
to the power of 112 as originally hoped. Triple

## Triple DES

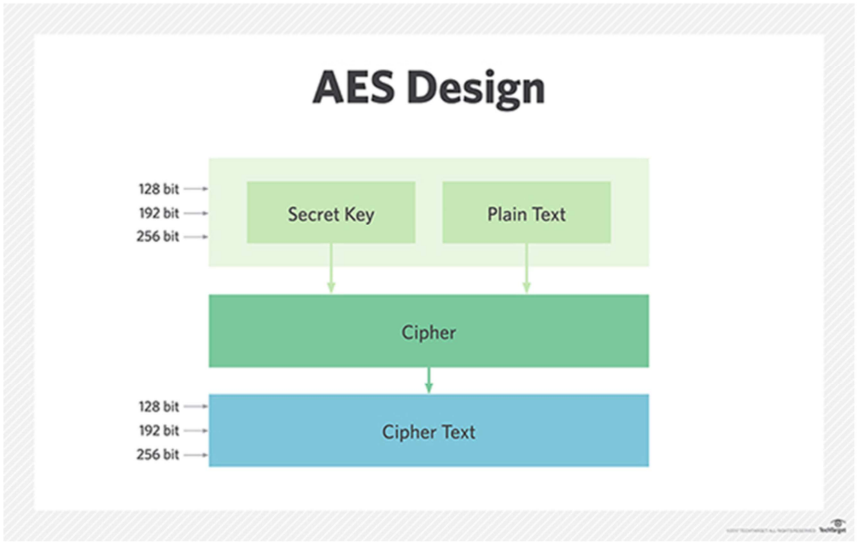



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The defeat of double DES resulted in the adoption of another improvement in how the DES algorithm could be modified to stand up better against brute force attacks. This improvement is known as Triple DES. Triple DES is much more secure, so much so that although attacks on it have been proposed, the data requirements of these have made them impractical. With Triple DES, there are three DES encryptions with either three or two different and separate keys that are used.

Managing three keys is more difficult, thus, many implementations will use the two-key method that reduces the key management requirement. The various ways of using Triple DES include the following:

- DES-EEE3: three DES encryptions with three different keys
- DES-EDE3: three DES operations in the sequence encrypt decrypt- encrypt with three different keys
- DES-EEE2 and DES-EDE2: same as the previous formats except that the first and third operations use the same key




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Reference https://searchsecurity.techtarget.com/definition/Advanced-Encryption-Standard

Please read this thoroughly and clear your concept on AES.

The Advanced Encryption Standard, or AES, is a symmetric block cipher chosen by the U.S. government to protect classified information and is implemented in software and hardware throughout the world to encrypt sensitive data.

The National Institute of Standards and Technology (NIST) started development of AES in 1997 when it announced the need for a successor algorithm for the Data Encryption Standard (DES), which was starting to become vulnerable to brute-force attacks.
This new, advanced encryption algorithm would be unclassified and had to be "capable of protecting sensitive government information well into the next century," according to the NIST announcement of the process for development of an advanced encryption standard algorithm. It was intended to be easy to implement in hardware and software, as well as in restricted environments (for example, in a smart card) and offer good defenses against various attack

techniques.

**AES features**

The selection process for this new symmetric key algorithm was fully open to public scrutiny and comment; this ensured a thorough, transparent analysis of the designs submitted.

NIST specified the new advanced encryption standard algorithm must be a block cipher capable of handling 128 bit blocks, using keys sized at 128, 192, and 256 bits; other criteria for being chosen as the next advanced encryption standard algorithm included:

**Security:** Competing algorithms were to be judged on their ability to resist attack, as compared to other submitted ciphers, though security strength was to be considered the most important factor in the competition.

**Cost:** Intended to be released under a global, nonexclusive and royalty-free basis, the candidate algorithms were to be evaluated on computational and memory efficiency.

**Implementation:** Algorithm and implementation characteristics to be evaluated included the flexibility of the algorithm; suitability of the algorithm to be implemented in hardware or software; and overall, relative simplicity of implementation.

**Choosing AES algorithms**

Fifteen competing symmetric key algorithm designs were subjected to preliminary analysis by the world cryptographic community, including the National Security Agency (NSA). In August 1999, NIST selected five algorithms for more extensive analysis. These were:

MARS, submitted by a large team from IBM Research

RC6, submitted by RSA Security

Rijndael, submitted by two Belgian cryptographers, Joan Daemen and Vincent Rijmen

Serpent, submitted by Ross Anderson, Eli Biham and Lars Knudsen

Twofish, submitted by a large team of researchers from Counterpane Internet Security, including noted cryptographer Bruce Schneier

Implementations of all of the above were tested extensively in ANSI C and Java languages for speed and reliability in encryption and decryption; key and algorithm setup time; and resistance to various attacks, both in hardware- and software-centric systems. Members of the global cryptographic community conducted detailed analyses (including some teams that tried to break their own submissions).

After much feedback, debate and analysis, the Rijndael cipher -- a mash of the Belgian creators' last names Daemen and Rijmen -- was selected as the proposed algorithm for AES in October 2000 and published by NIST as U.S. FIPS PUB 197. The Advanced Encryption Standard became effective as a federal government standard in 2002. It is also included in the International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) 18033-3 standard, which

specifies block ciphers for the purpose of data confidentiality.
In June 2003, the U.S. government announced that AES could be used to protect classified information, and it soon became the default encryption algorithm for protecting classified information as well as the first publicly accessible and open cipher approved by the NSA for top-secret information. The NSA chose AES as one of the cryptographic algorithms to be used by its Information Assurance Directorate to protect national security systems.
Its successful use by the U.S. government led to widespread use in the private sector, leading AES to become the most popular algorithm used in symmetric key cryptography. The transparent selection process helped create a high level of confidence in AES among security and cryptography experts. AES is more secure than its predecessors -- DES and 3DES -- as the algorithm is stronger and uses longer key lengths. It also enables faster encryption than DES and 3DES, making it ideal for software applications, firmware and hardware that require either low latency or high throughput, such as firewalls and routers. It is used in many protocols such as Secure Sockets Layer (SSL)/Transport Layer Security (TLS) and can be found in most modern applications and devices that need encryption functionality.

**How AES encryption works**

AES comprises three block ciphers: AES-128, AES-192 and AES-256. Each cipher encrypts and decrypts data in blocks of 128 bits using cryptographic keys of 128-, 192- and 256-bits, respectively. The Rijndael cipher was designed to accept additional block sizes and key lengths, but for AES, those functions were not adopted.
Symmetric (also known as secret-key) ciphers use the same key for encrypting and decrypting, so the sender and the receiver must both know -- and use -- the same secret key. All key lengths are deemed sufficient to protect classified information up to the "Secret" level with "Top Secret" information requiring either 192- or 256-bit key lengths. There are 10 rounds for 128-bit keys, 12 rounds for 192-bit keys and 14 rounds for 256-bit keys -- a round consists of several processing steps that include substitution, transposition and mixing of the input plaintext and transform it into the final output of ciphertext.
The AES encryption algorithm defines a number of transformations that are to be performed on data stored in an array. The first step of the cipher is to put the data into an array; after which the cipher transformations are repeated over a number of encryption rounds. The number of rounds is determined by the key length, with 10 rounds for 128-bit keys, 12 rounds for 192-bit keys and 14 rounds for 256-bit keys.
The first transformation in the AES encryption cipher is substitution of data using a substitution table; the second transformation shifts data rows, the third mixes columns. The last transformation is a simple exclusive or (XOR) operation performed on each column using a different part of the encryption key -- longer keys need more rounds to complete.
AES encryption transforms array data by shuffling rows and columns, and substitutions based

on the encryption key.

**Attacks on AES encryption**

Research into attacks on AES encryption has continued since the standard was finalized in 2000. Various researchers have published attacks against reduced-round versions of the Advanced Encryption Standard.

Margaret Rouse asks:

**How could the selection process used by NIST for the Advanced Encryption Standard be improved?**

**Join the Discussion**

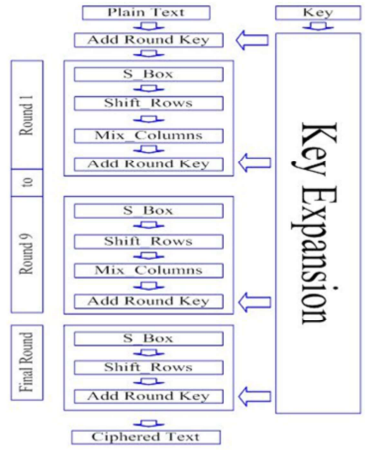
In 2005, cryptographer Daniel J. Bernstein published a paper, "Cache-timing attacks on AES," in which he demonstrated a timing attack on AES capable of achieving a "complete AES key recovery from known-plaintext timings of a network server on another computer."

A research paper published in 2011, titled "Biclique Cryptanalysis of the Full AES," by researchers Andrey Bogdanov, Dmitry Khovratovich, and Christian Rechberger, demonstrated that by using a technique called a biclique attack, they could recover AES keys faster than a brute-force attack by a factor of between three and five, depending on the cipher version. However, even this attack does not threaten the practical use of AES due to its high-computational complexity.

AES has proven to be a reliable cipher, and the only practical successful attacks against AES have leveraged side-channel attacks on weaknesses found in the implementation or key management of specific AES-based encryption products. Side-channel attacks exploit flaws in the way a cipher has been implemented rather than brute force or theoretical weaknesses in a cipher. The Browser Exploit Against SSL/TLS (BEAST) browser exploit against the TLS v1.0 protocol is a good example; TLS can use AES to encrypt data, but due to the information that TLS exposes, attackers managed to predict the initialization vector block used at the start of the encryption process.

# Rijndael

- Developed by two Belgian cryptographers where it's based on SP-Networks.
- The ciphering process is divided into three stages:
  1) Key Expansion
  2) Rounds
  3) Final Round




RIJNDAEL BLOCK DIAGRAM

C. Chitu and M. Giesner, An FPGA implementation of the AES-Rijndael in OCB/ECB modes of operation, Microelectronics Journal 36 (2005) 139146,21 October, 2004.

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6

Reference https://searchsecurity.techtarget.com/definition/Rijndael

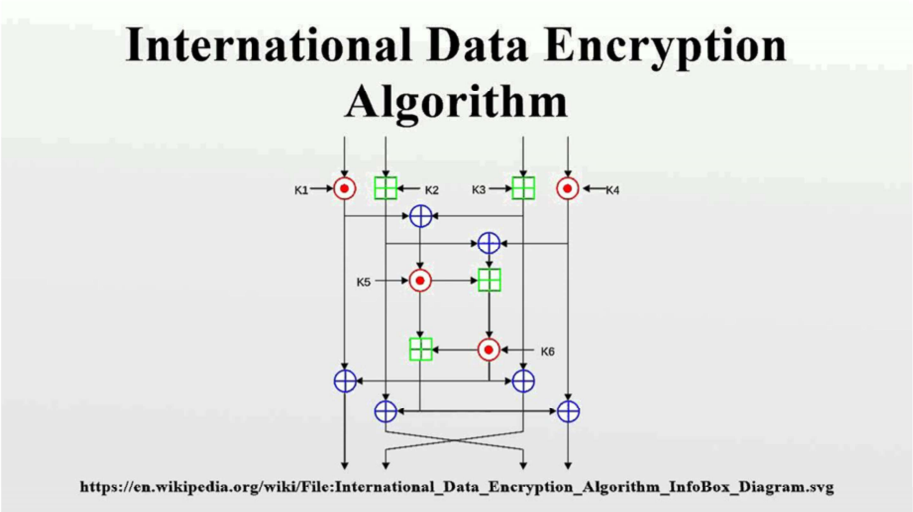
Rijndael (pronounced rain-dahl) is the algorithm that has been selected by the U.S. National Institute of Standards and Technology (NIST) as the candidate for the Advanced Encryption Standard (AES). It was selected from a list of five finalists, that were themselves selected from an original list of more than 15 submissions. Rijndael will begin to supplant the Data Encryption Standard (DES) - and later Triple DES - over the next few years in many cryptography applications. The algorithm was designed by two Belgian cryptologists, Vincent Rijmen and Joan Daemen, whose surnames are reflected in the cipher's name. Rijndael has its origins in Square, an earlier collaboration between the two cryptologists.
The Rijndael algorithm is a new generation symmetric block cipher that supports key sizes of 128, 192 and 256 bits, with data handled in 128-bit blocks - however, in excess of AES design criteria, the block sizes can mirror those of the keys. Rijndael uses a variable number of rounds, depending on key/block sizes, as follows:
9 rounds if the key/block size is 128 bits
11 rounds if the key/block size is 192 bits

13 rounds if the key/block size is 256 bits
Rijndael is a substitution linear transformation cipher, not requiring a Feistel network. It use triple discreet invertible uniform transformations (layers). Specifically, these are: Linear Mix Transform; Non-linear Transform and Key Addition Transform. Even before the first round, a simple key addition layer is performed, which adds to security. Thereafter, there are Nr-1 rounds and then the final round. The transformations form a State when started but before completion of the entire process.
The State can be thought of as an array, structured with 4 rows and the column number being the block length divided by bit length (for example, divided by 32). The cipher key similarly is an array with 4 rows, but the key length divided by 32 to give the number of columns. The blocks can be interpreted as unidimensional arrays of 4-byte vectors.
The exact transformations occur as follows: the byte subtransformation is nonlinear and operates on each of the State bytes independently - the invertible S-box (substitution table) is made up of 2 transformations. The shiftrow transformation sees the State shifted over variable offsets. The shift offset values are dependent on the block length of the State. The mixcolumn transformation sees the State columns take on polynomial characteristics over a Galois Field values (28), multiplied x4 + 1 (modulo) with a fixed polynomial. Finally, the roundkey transform is XORed to the State. The key schedule helps the cipher key determine the round keys through key expansion and round selection.
Overall, the structure of Rijndael displays a high degree of modular design, which should make modification to counter any attack developed in the future much simpler than with past algorithm designs.

**Was the best choice really Rijndael?**

Our Cryptography expert, Borys Pawliw comments: "The AES selection was always going to be a compromise, balancing various factors such as overall security, performance, and efficiency. As such, it was unlikely that the selection of any one algorithm would receive unanimous praise from all quarters. Rijndael's selection has been criticized by some because the algorithm does not appear to be as secure as some of the other choices.This criticism is valid theoretically, but does not mean that data secured using this algorithm is going to be unacceptably vulnerable to attack. Although Rijndael may not have been the most secure algorithm from an academic viewpoint, defenders claim that it is more than likely secure enough for all applications in the real world and can be enhanced by simply adding more rounds. Attacks on the algorithm have succeeded only in an extremely limited environment and, while interesting from a mathematical viewpoint, appear to have little consequence in the real world."

## International Data Encryption Algorithm

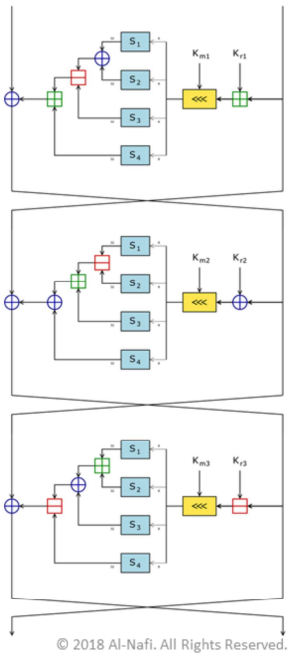



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Reference https://searchsecurity.techtarget.com/definition/International-Data-Encryption-Algorithm

DEA (International Data Encryption Algorithm) is an encryption algorithm developed at ETH in Zurich, Switzerland. It uses a block cipher with a 128-bit key, and is generally considered to be very secure. It is considered among the best publicly known algorithms. In the several years that it has been in use, no practical attacks on it have been published despite of a number of attempts to find some. IDEA is patented in the United States and in most of the European countries. The patent is held by Ascom-Tech. Non-commercial use of IDEA is free. Commercial licenses can be obtained by contacting Ascom-Tech.

# CAST




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8

CAST was developed in 1996 by Carlisle Adams and Stafford Tavares. CAST-128 can use keys between 40 and 128 bits in length and will do between 12 and 16 rounds of operations related to substitutions and transpositions, depending on key length. CAST-128 is a Feistel-type block cipher with 64-bit blocks. CAST-256 was submitted as an unsuccessful candidate for the AES competition. CAST-256 operates on 128-bit blocks and with keys of 128, 192, 160, 224, and 256 bits. It performs 48 rounds and is described in RFC 2612.

# Secure and Fast Encryption Routine (SAFER)

## SAFER+ Block Ciphers

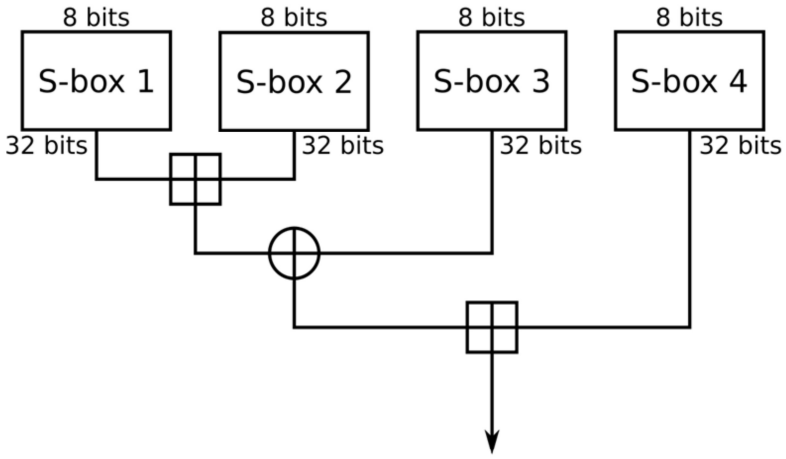
- To Authenticate Bluetooth device
- An enhancement of SAFER (Secure And Fast Encryption Routine)
- A Fiestel cipher with a 128-bit block size
- Two components:
  - Key scheduling component
  - Encryption component
    - Eight identical rounds (two subkeys for each round)
    - An output transformation (one subkey)

J. Wang and Z. Kissel. Introduction to Computer Network Security: Theory and Practice. Wiley 2015

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All of the algorithms in SAFER are patent-free. The algorithms were developed by James Massey and work on either 64-bit input blocks (SAFER-SK64) or 128-bit blocks (SAFER-SK128). A variation of SAFER is used as a block cipher in Bluetooth.

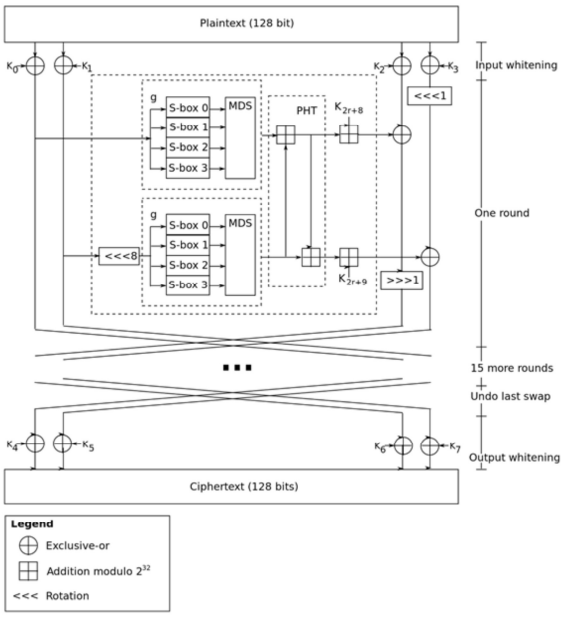
## Blowfish Encryption




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Blowfish is another example of a symmetric algorithm developed by Bruce Schneier. It is considered to be an extremely fast cipher, and one of its extremely useful advantages is that it requires very little system memory. It is also a Feistel-type cipher in that it divides the input blocks into two halves and then uses them in XORs against each other. However, it varies from the traditional Feistel ciphers in that Blowfish does work against both halves, not just one. The Blowfish algorithm operates with variable key sizes, from 32 up to 448 bits on 64-bit input and output blocks.

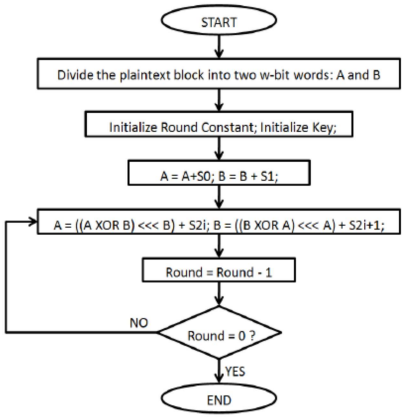
# Twofish Encryption




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Twofish was one of the finalists for the AES competition mentioned earlier. It is an adapted version of Blowfish developed by a team of cryptographers led by Bruce Schneier. It can operate with keys of 128, 192, or 256 bits on blocks of 128 bits. Just like DES, it performs 16 rounds during the encryption and decryption process.

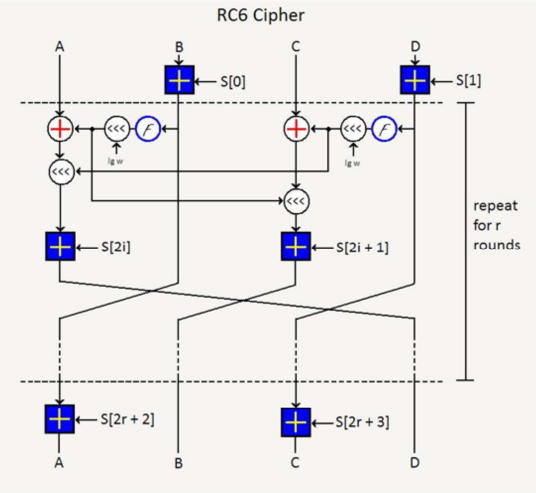
# Rivest Cipher 5 (RC5)




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RC5 is a fast block cipher designed by Ron Rivest. The algorithm was designed to be used in existing security products and in a number of internet protocols. It was explicitly designed to be simple to implement in software, therefore, the algorithm does not support any type of bit permutations. Rivest designed a lengthy sub-key generation phase into the algorithm to make brute force key searching substantially more difficult without slowing down conventional one-key uses of RC5. Today's RC5 is a parameterized algorithm with a variable block size, a variable key size, and a variable number of rounds. Allowable choices for the block size are 32, 64, and 128 bits. The number of rounds can range anywhere from 0 to 255, while the key can range from 0 bits to 2040 bits in size. There are three routines in RC5: key expansion, encryption, and decryption. In the key expansion routine, the user-provided secret key is expanded to fill a key table whose size depends on the number of rounds. The key table is then used in both encryption and decryption. The encryption routine consists of three primitive operations: integer addition, bitwise XOR, and variable rotation.

# Rivest Cipher 6 (RC6)




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RC6 is a block cipher based on RC5 and, just like its predecessor, it is a variable parameterized algorithm where the block size, the key size, and the number of rounds are variable. The upper limit on the key size is 2040 bits, which experts agree should certainly make it strong for quite a few years. When RC6 was designed, they built two new features in it compared to RC5. The first is the inclusion of integer multiplication and the use of four 4-bit working registers instead of two 2-bit working registers. Integer multiplication is used to increase the diffusion achieved per round so that fewer rounds are needed and the speed of the cipher can be increased.

# Rivest Cipher 4 (RC4)

## RC4

- Was developed by Ron Rivest in 1987
- "Rivest Cipher 4"
- Most widely used stream cipher
- Used in i.e. SSL, WEP, WPA and TLS
- Was secret until 1994 when it was leaked

```
for i from 0 to 255
   S[i] := i
endfor
l := keylength
j := 0
for i from 0 to 255
   j := (j + S[i] + key[i mod l])
   mod 256 swap(&S[i],&S[j])
endfor
i := 0
j := 0
while GeneratingOutput:
   i := (i + 1) mod 256
   j := (j + S[i]) mod 256
   swap(&S[i],&S[j])
   output S[(S[i]+S[j]) mod 256]
endwhile
```

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RC4, a stream-based cipher, was developed in 1987 by Ron Rivest for RSA Data Security and has become the most widely used stream cipher, being deployed, for example, in WEP and Secure Socket Layer/Transport Layer Security (SSL/TLS). RC4 can use a variable length key ranging from 8 to 2,048 bits (1 to 256 bytes) and a period of greater than 10,100. This means that in implementations, it is possible to ensure that the keystream should not repeat for at least that length. If RC4 is used with a key length of at least 128 bits, there are currently no practical ways to attack it. Confusion exists in the industry as to the weakness in WEP in regards to WEP using RC4 and its weakness. The published successful attacks against the use of RC4 in WEP applications are actually related to problems with the implementation of the algorithm, not the algorithm itself.